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# Research to real-world impact: evidence-based knowledge mobilization to improve support for UK women veterans who have experienced military sexual trauma

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## ABSTRACT

**Background:** Military sexual trauma (MST) disproportionately impacts women veterans globally. UK research highlights that these in-service experiences may contribute to unique mental and physical health needs. Yet, service providers are often perceived to be unaware of and unable to meet these needs.

**Objective:** To evaluate a national training programme that synthesised evidence to support UK women veterans. Situated within a wider knowledge mobilization (KMb) initiative and guided by the Knowledge-to-Action (KTA) framework, a three-part programme was developed for different professional audiences to enhance understanding of and support for women veterans.

**Method:** Co-produced and grounded in KMb theory. Course 1 aimed to enhance knowledge of women veterans' health needs. Course 2 targeted healthcare professionals and veteran specialist providers, exploring evidence-based considerations for promoting service engagement. Course 3 aimed to increase UK clinical capacity in the best-evidenced treatment for post-traumatic stress disorder associated with MST in women veterans, through training mental health professionals in cognitive processing therapy (CPT). Mixed-methods data were collected from all programme attendees. CPT attendees completed pre- and 3-month post-training measures; quantitative data were analysed using a Wilcoxon signed-rank test and qualitative data using manifest content analysis.

**Results:** Course 1 was completed by 526 individuals, 363 completed Course 2, and 98 completed Course 3. Course 3 attendees were split across two cohorts ($n$ = 57; $n$ = 41). Compared with pre-training, 3-month post-training self-efficacy in delivering CPT across cohort one increased significantly ($p$ < .001, $r$ = 0.915). Content analysis identified three themes: (1) Clinician confidence delivering CPT; (2) Clinicians' perceptions of CPT; and (3) CPT in a service system.

**Conclusions:** The education programme addressed gaps in UK service provider knowledge and increased CPT clinical capacity. The wider KMb initiative demonstrates how research can be translated into impact for military populations and highlights the value of co-produced KMb approaches.

## ARTICLE HISTORY

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## KEYWORDS

Knowledge mobilization; capacity building; military sexual trauma; women veterans; PTSD; trauma; military

## PALABRAS CLAVE

Movilización de conocimiento; fortalecimiento de capacidades; trauma sexual militar; mujeres veteranas; TEPT; trauma; militar

## HIGHLIGHTS

- Military sexual trauma (MST) is a potentially leading contributor to military-service linked post-traumatic stress disorder (PTSD) among women who have served in the armed forces.
- Guided by knowledge mobilization theory, this study used the international evidence base to enhance service support for UK women veterans affected by MST.
- A three-part online education programme was co-produced with women veterans and aimed at different UK professional groups to help them better support those affected by MST.

## Investigación sobre impacto en el mundo real: movilización de conocimiento basado en la evidencia para mejorar el apoyo a las veteranas del Reino Unido que han sufrido trauma sexual militar

**Antecedentes:** El trauma sexual militar (MST en su sigla en inglés) afecta de forma desproporcionada a las mujeres veteranas a nivel mundial. La investigación en el Reino Unido destaca que estas experiencias en servicio pueden contribuir a un perfil único de las necesidades de salud mental y física de las mujeres veteranas. Sin embargo, a menudo se percibe que los proveedores de servicios y los responsables políticos desconocen estas necesidades y son incapaces de satisfacerlas.

**Objetivo:** Evaluar un programa nacional de formación para fortalecer la capacidad de apoyo a las mujeres veteranas en el Reino Unido. Dentro de una iniciativa más amplia de movilización del conocimiento (KMb en su sigla en inglés), se desarrolló un programa de tres partes para diferentes públicos profesionales con el fin de mejorar la comprensión y el apoyo a las mujeres veteranas del Reino Unido.

**Método:** Coproducido y basado en la teoría KMb, el programa sintetizó evidencia internacional. El Primer Curso tenía como objetivo ampliar el conocimiento general sobre las necesidades de salud mental de las mujeres veteranas. El Curso Dos se dirigió a profesionales sanitarios y proveedores especialistas veteranos e incluyó consideraciones basadas en la evidencia para promover la participación en el servicio. El Curso Tres tenía

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como objetivo aumentar la capacidad clínica del Reino Unido en el tratamiento mejor evidenciado para el TEPT asociado al MST en mujeres veteranas, mediante la formación de profesionales de la salud mental en Terapia de Procesamiento Cognitivo (CPT en su sigla en inglés). Se recogieron datos de métodos mixtos de todos los asistentes al curso. Los asistentes al CPT también completaron las medidas previas y tres meses después de la formación; los datos cuantitativos se analizaron mediante una prueba de rango por signos de Wilcoxon, y los datos cualitativos mediante análisis de contenido manifiesto.
**Resultados:** El Curso Uno fue completado por 457 personas, 322 completaron el Curso Dos y 57 completaron el Curso Tres. En comparación con el preentrenamiento, la autoeficacia en la entrega de CPT después del entrenamiento aumentó significativamente ($p < .001$, $r = 0.915$). El análisis de contenido de las respuestas post-entrenamiento CPT identificó tres temas: 1) Confianza del clínico en la impartición de CPT; 2) Percepciones de los clínicos sobre el CPT; y 3) CPT en un sistema de servicios.
**Conclusiones:** El programa educativo pareció abordar las lagunas en el conocimiento de los proveedores de servicios en el Reino Unido y aumentar la capacidad clínica de la CPT. La iniciativa más amplia de los KMb demuestra cómo la investigación puede traducirse en impacto para las poblaciones militares y destaca el valor de los enfoques de KMb coproducidos.

## 1. Background

Military sexual trauma (MST) is a continuum of violence from sexual harassment and bullying to sexual assault, perpetrated by serving personnel on other service members during military service (Herriott et al. 2024). Although there is currently no universally accepted definition of MST, it has been found to disproportionately impact women across military health research globally (Buyse et al., 2023; Moreau et al., 2021; Mota et al., 2023). A US meta-analysis found that 38.4% of serving and formerly serving women reported having experienced MST, over nine times higher than that reported by men (3.9%) (Wilson, 2018). While both men and women report experiencing MST, the higher number of serving men means that absolute numbers may be comparable, although underreporting remains a prevailing issue for all (Ministry of Defence, 2019, 2021). MST has been associated with a plethora of negative health outcomes, including physical pain (Shapiro et al., 2023), gastrointestinal problems (Pulverman & Creech, 2021), and symptoms of anxiety and depression (MacDermid et al., 2025). In women veterans specifically, MST has been identified as a potentially leading contributor to military-service linked post-traumatic stress disorder (PTSD) (Campbell & Murphy, 2023; Surís et al., 2004).

Despite the commonalities evidenced across different militaries, the world's armed forces are not homogeneous. Therefore, it is necessary to understand the impacts of MST within specific national veteran populations (Godier-McBard et al., 2021). In the UK, a veteran is defined someone who has completed at least one full day in the armed forces and is no longer serving (Burdett et al., 2012). UK veteran health research has predominantly focused on the male veteran majority, although there are estimated to be over 250,000 women veterans, representing approximately 13% of the total veteran population (National Records of Scotland, 2024; Office for National Statistics, 2021). With a target to reach 30% inflow of women to the military by 2030, the number of women veterans in future years is expected to increase, alongside the need to better understand and support this group through and beyond military service (Ministry of Defence, 2022). With international evidence indicating that MST is likely to be a significant contributor to poor mental health in women veterans, there is a need to build evidence-based interventions and support provision for UK women who have experienced MST.

### 1.1. Moving from research to action

Evidence-based practice integrates research, clinical expertise, and contextual understanding of patient groups to optimize outcomes through robust, consistent, and cost-effective care (Cook et al., 2017). This requires the transfer of research findings into practice, variously referred to as knowledge transfer, implementation science, and knowledge exchange, to name a few. There are myriad extant models to describe this process of 'closing the gap' between research and its real-world operationalization and implementation (Davies et al., 2015; Strifler et al., 2018). Current knowledge mobilization (KMb) (the authors' preferred term) models emphasize a non-linear set of stages that embed research evidence into practice within healthcare and societal 'systems' (Durrant et al., 2024).

KMb approaches are unstandardized and iterative in nature, although core functions include 'dissemination and engagement of academic research beyond that made possible by traditional forms of scholarly communication' (Phipps, 2011) and capacity building, which serves to augment the capability of organizations to achieve and sustain longer term objectives (Bate et al., 2004). Co-production and co-design are

also commonly used to bridge the knowledge gap, enhance acceptability among policy makers, increase relevance to target populations, and improve the likelihood of successful implementation within service systems (Grindell et al., 2022).

The most widely used KMb model is Graham and colleagues' Knowledge-to-Action (KTA) framework (Field et al., 2014; Graham et al., 2006), which divides the process into two linked concepts (Figure 1). The first concept – knowledge creation – is concerned with tailoring and refining information to be of increasing utility. This encompasses three stages: gathering knowledge from primary research ('knowledge inquiry'); interpretation of primary research through syntheses such as systematic reviews ('knowledge synthesis'); and finally, presenting this synthesized knowledge in accessible, audience-appropriate formats to promote evidence-based decision making ('knowledge tools and products'). The second concept – the action cycle – details the activities required to implementation this knowledge. This is split into seven stages: identify a problem that needs addressing, and identify, review, and select the knowledge or research relevant to the problem; adapt the identified knowledge to the local context; assess barriers to using the knowledge; select, tailor, and implement interventions to promote the use of knowledge; monitor knowledge use; evaluate the outcomes of using the knowledge; and sustain ongoing knowledge use. Although the KTA framework describes two concepts and constituent stages, the boundaries between each are seen as fluid, and action phases need not be sequential or discrete.

This paper examines how the KTA framework was applied to a series of interlinked projects focusing on improving support for UK women veterans who have experienced MST. It first outlines in brief the knowledge creation stage, before detailing the application of the action cycle in designing, implementing, and assessing interventions to help address the impact of MST through KMb.

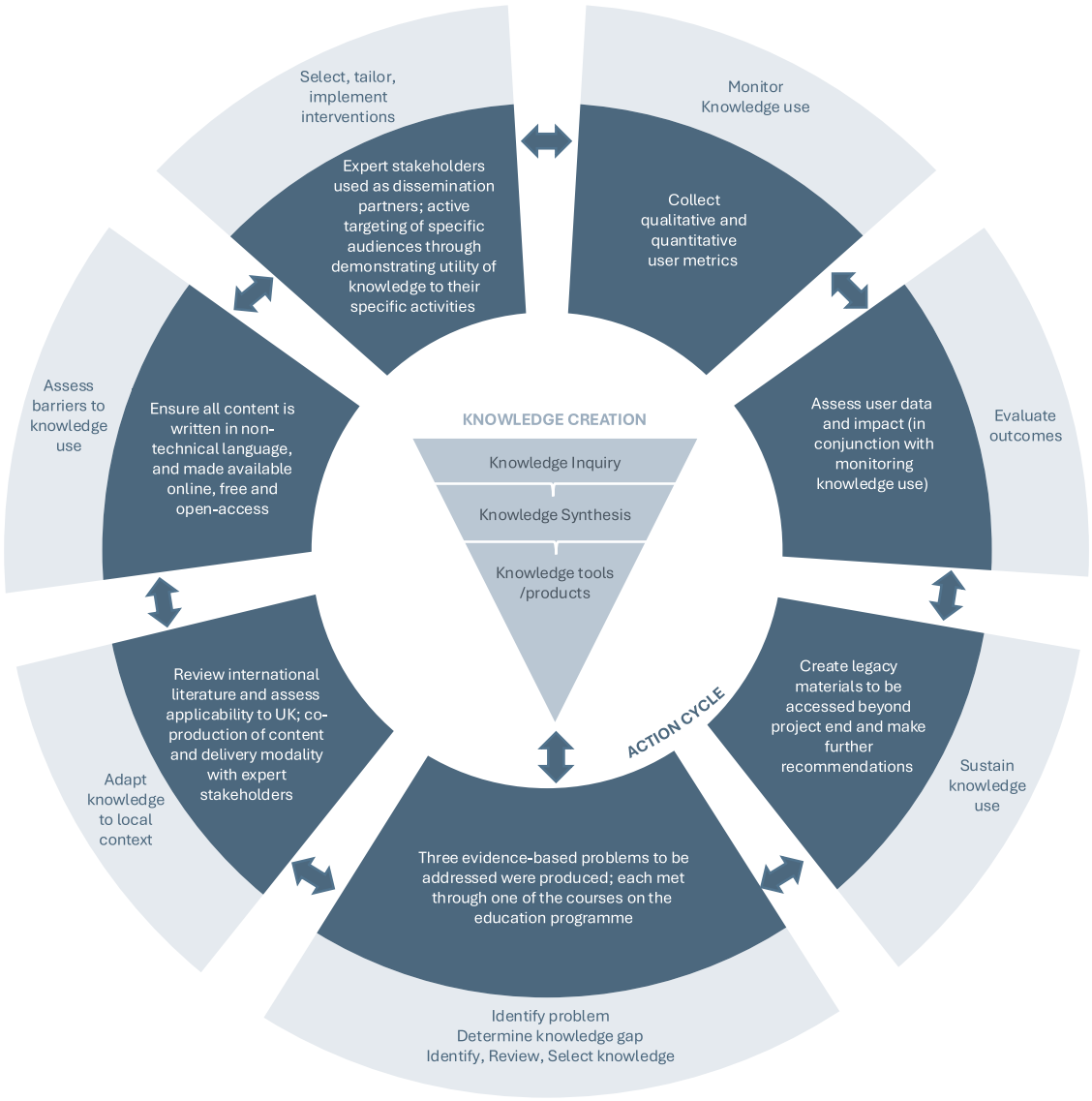



**Figure 1.** Knowledge-to-Action (KTA) framework stages and action taken.

#### 1.1.1. Knowledge inquiry: building the evidence base around MST

A cohort of 750 UK women veterans was recruited in the largest study to date exploring their experiences during military service and their current health profile; for full methodology, see Hendrikx, Williamson, et al. (2021). The study identified a high prevalence of common mental health disorders (CMDs) (28.6%) and probable PTSD (10.8%) in the cohort. In addition, 23% reported having experienced sexual harassment, 23% emotional bullying, and 5% sexual abuse (Hendrikx, Williamson, & Murphy, 2021), figures broadly comparable to other countries (Wilson, 2018). Significant associations were found between reporting a high number of adverse childhood experiences and experiencing MST (Williamson et al., 2022). Crucially, MST was found to be significantly associated with increased odds of reporting probable PTSD as a veteran (Hendrikx, Williamson, & Murphy, 2021). Furthermore, using a checklist of in-service challenging experiences co-produced with women veterans and subject-matter experts, 42.7% of the cohort reported that sexist assumptions were made about them during their service and 40.1% felt that they had to be better than their male colleagues to receive the same recognition at work. These in-service challenges were also significantly associated with an increased risk of reporting poor mental and physical health as a veteran (Baumann et al., 2022).

Additional analysis explored the impact of past gender-based discrimination, such as forced discharge for pregnancy and experiences of the so-called 'gay ban' (Biscoe et al., 2024). These studies cumulatively established an evidence base, which indicated that UK women veterans experienced idiosyncratic, gender-based challenges during their military service that were different from those of their male counterparts, and which were likely to contribute to poor mental health as a veteran. MST specifically was reported as a significant contributor. Accordingly, the need to focus on MST as significant potential in-service adversity with long-term health implications for UK women veterans was indicated.

#### 1.1.2. Knowledge synthesis: exploring women veterans' needs and experiences

Further exploration of MST and resultant PTSD in UK women veterans was conducted. In line with the KTA framework, research questions were formulated specifically relating to how support services could be improved for women veterans. These focused on what evidence-based treatments work and where systemic improvements could be made. A systematic review of treatments specifically for PTSD related to MST in women veterans found that cognitive processing therapy (CPT) was the best-evidenced current treatment (Campbell, Biscoe, et al., 2024). CPT is also a reccomended trauma focused cognitive behavioural therapy under the National Institute for Health and Care Excellence (NICE) guidelines (NICE, 2018). Furthermore, a subcohort of UK women veterans from the first study with scores indicative of probable PTSD was invited to take part in semi-structured interviews exploring their experiences of accessing psychological healthcare. Qualitative analysis showed that women veterans reported facing similar barriers to men veterans, such as stigma around help-seeking (Randles & Finnegan, 2022), but that these were exacerbated as the result of their minority status (Campbell et al., 2025; Campbell, Perry, et al., 2024). Additional salient barriers and facilitators included a perceived lack of professional knowledge and awareness about women veterans' needs and experiences, the invalidation of specific service traumatic experiences such as MST, and gender discrimination in the design, provision, and focus of veteran-specific services, which were viewed as catering for the male majority (Campbell et al., 2025).

#### 1.1.3. Knowledge tools and products: evidence-based recommendations for action

Drawing on the findings from the previous stages, core recommendations were co-produced with expert stakeholders (veteran experts, healthcare professionals, and women veterans with lived experience) to guide future improvements in support for UK women veterans, and specifically those who had experienced MST (Campbell & Murphy, 2023). Policy experts were also engaged to ensure that recommendations were applicable in the policy domain. These recommendations included:

- Increase healthcare professionals' awareness of women veterans, their military service experiences, and mental healthcare needs.
- Address specific barriers to engagement for women veterans within services, potentially through bespoke service design, access, and treatment pathways.
- Ensure that the best evidence-based treatment is available for women veterans who have experienced MST.

#### 1.1.4. The action cycle: the current study

The current study, guided by the KTA framework, aimed to address each of the three evidence-based recommendations by mobilizing knowledge to improve access to, and provision of, evidenced treatment and support for UK women veterans who have experienced MST.

A three-tiered education and training programme was developed as part of this action cycle. Course 1 aimed to enhance general knowledge about women

veterans' mental health needs, and was aimed at anyone who may have professional contact with women veterans. Course 2 targeted healthcare professionals and veteran specialist providers, and included evidence-based considerations for promoting service engagement. Since women veterans often seek support for mental health difficulties from non-veteran-specific services, engaging these groups was prioritized (Campbell & Murphy, 2023; Hooks et al., 2023). In line with KMb theory, to sustain the longer-term objective of the programme and ensure that best-evidenced support is available to women veterans in the UK, Course 3 aimed to increase UK clinical capacity in the best-evidenced treatment for PTSD associated with MST in women veterans by providing fully funded training places, for appropriately qualified mental health professionals, in CPT (Campbell, Biscoe, et al., 2024).

## 2. Method

### 2.1. Overview

The KTA framework's action cycle was used as a guide (Figure 1) throughout the KMb process. The three evidence-based recommendations for action were reframed as problems against which the evidence base could be mobilized to provide a solution. In response, an online three-tiered education and training programme was conceived, with each course responding to one problem, and aimed at a specific professional end-user group. These courses comprised:

(1) an introductory course designed to raise basic awareness of woman veterans' needs and experiences, aimed at anybody who had professional contact with women veterans
(2) an advanced course designed to explore how services can be adapted to be more accessible, welcoming, and appropriate to women veterans, aimed at those working in veteran-specific services and healthcare providers
(3) training in CPT to increase UK clinical capacity in the best-evidenced treatment for PTSD related to MST in women veterans, for mental health professionals who work with women veterans.

To reduce barriers to access, the programme was made freely available via an online open access website (https://veterans.training).

### 2.2. Co-production stakeholder group

The programme content and structure were co-produced with a stakeholder group comprised of women veterans, currently serving military personnel, and experts including clinical, academic, and support practitioners. The group included 12 policy and practice partners from the UK ($n = 7$), Australia ($n = 2$), Canada ($n = 2$), and New Zealand ($n = 1$). The stakeholder group was asked to anonymously prioritize topics for inclusion in the introductory and advanced courses, as well as providing recommendations on delivery modality (e.g. self-directed versus live seminar learning) and terminology preferences (e.g. the acceptability of the term 'MST'). Stakeholders also completed a semi-structured questionnaire following course testing, with feedback on usability, acceptability, design, and content.

### 2.3. Course content

#### 2.3.1. Course 1: Introduction to women veterans' experiences and needs

Course 1 was a 10 min, online, self-guided, foundational knowledge course. It explored the unique experiences of UK women veterans, challenges faced during military service, global conceptualizations of MST, and issues related to women veteran's mental health and healthcare. Each topic included a brief true-or-false quiz to consolidate learning. Four true-or-false questions were asked following each topic, resulting in a total of 12 questions across Course 1. True-or-false quizzes were chosen because this method of consolidation provides a frictionless method of engaging participants across learning materials (Uner et al., 2022), compatible with the WordPress content management system used to host the online education programme. Courses 2 and 3 were both advertised on completion, and all attendees confirmed their consent to be contacted and provided informed consent for their data to be used under the privacy policy.

#### 2.3.2. Course 2: Advanced understanding of women veterans – how services can better meet their needs

Course 2 was a 2 h live online workshop, facilitated by members of the project team (DD and GMC), focusing on addressing barriers to engagement with support services among women veterans by enhancing knowledge among veteran specialist providers and developers (e.g. veteran charities) and non-specialist healthcare providers (e.g. family doctors). The workshop contextualized and summarized the evidence base, explored women veterans' experiences accessing healthcare services, and presented evidence-based considerations for promoting service engagement. During workshops, attendees were encouraged to engage in discussions on the material covered and share their own professional experiences of working in the sector.

#### 2.3.3. Course 3: Advanced understanding of women veterans plus cognitive processing therapy (CPT) training

Course 3 consisted of a truncated, 1 h long version of Course 2 plus foundational CPT training, delivered online over 2 days to appropriately qualified professionals working with veterans. Course material included a series of training videos and written content, developed and delivered by two clinical experts in CPT. Training covered the theory underlying CPT, session-by-session review of the CPT protocol with examples specific to MST, common challenges to implementation, and strategies to overcome challenges, such as clinical adherence and managing comorbidities (Resick et al., 2024).

### 2.4. Dissemination

The programme was disseminated using a multi-pathway outreach strategy. This included targeted promotion via social media platforms, network dissemination from the stakeholder group, and direct contact with relevant organizations, including UK military charities, healthcare professional bodies, and veteran-focused employers. Prospective audiences were identified based on their professional relevance to the programme's objectives; outreach was conducted primarily through e-mail. Prospective audiences were guided to the website. All users were requested to complete Course 1 and then register their subsequent interest in Course 2 or Course 3. Those who indicated an interest in Course 2 were invited to attend by e-mail. Appropriately qualified mental health professionals who registered their interest in Course 3 were invited to attend the CPT training.

### 2.5. Data collection and analysis

Qualitative measures of participant demographics, location, and occupational details were recorded at sign-up for all courses. A total of 681 individuals signed up to Course 1, of whom 526 completed the full course and provided response data, representing an attrition rate of 22.76%.

For Course 2, 0.55% demographic data were recorded as 'unknown' owing to participants attending the live online sessions anonymously. Missing demographic data were also recorded for Course 3 (5%), again categorized as 'unknown'. Qualitative and quantitative analyses were conducted using available data only. Attendees' qualitative feedback was collected at the end of all courses via semi-structured online questionnaires, assessing the appropriateness of each course content and ongoing applicability to attendees' professional role (Appendix A).

Qualitative feedback from the post-training survey in Course 3 additionally focused on factors influencing CPT implementation. Responses were analysed using a manifest content analysis framework, chosen for its flexibility as a deductive or inductive analytical approach and its suitability for use with open-ended questionnaire data (Bengtsson, 2016). A manifest approach was used to explore clinicians' surface-level perspectives in response to the questions asked; manifest content analysis is concerned with 'What has been said?' and during data analysis stays close to the text, often using the participants' words to describe what is visible to the researcher (Berg, 2002). Response data were read for familiarization, and meaning units were extracted, condensed, and coded inductively, whereby the coding list was created during data analysis. A code list was created, and frequently occurring codes or related codes were organized into subthemes, which were organized into main themes. Categorization modifications were made subject to peer consultation (DD and GMC). The resulting themes and subthemes were discussed among the research team for refinement before consensus was reached (Table 2).

In addition, self-efficacy in delivering CPT for trainees on Course 3 was assessed pre-training and 3 months post-training, using a modified item from the therapist's self-efficacy measure by Pace et al. (2021). Responses were collected via an online semi-structured questionnaire. Trainees were asked 'How confident are you currently delivering CPT?' Responses were scored on a four-point scale: Not at all confident (1), A little confident (2), Quite confident (3), Very confident (4). Quantitative data were analysed using a Wilcoxon signed-rank test, whereas qualitative data were analysed using manifest content analysis (Bengtsson, 2016). Descriptive statistics were reported for all demographic and outcome measures as appropriate, and across time-points. Analysis was performed using IBM SPSS 29.0 (IBM Corp., Armonk, NY, USA).

## 3. Results

### 3.1. Course 1: Introduction to women veterans' experiences and needs

A total of 526 professionals completed Course 1. Attendees represented a range of service organizations and sectors, almost one-third had served in the military (30.03%), most worked with armed forces veterans or for an organization working with veterans (96.76%), and 192 attendees (36.50%) were registered mental health professionals. The majority (79.28%) were from England; numbers across Wales (7.98%) and Northern Ireland (2.85%) were representative of the UK population, although attendance from Scotland was lower than anticipated (8.37%). Mean quiz percentage correctness showed a progressive increase

across the three quizzes (Quiz 1: 76.7%; Quiz 2: 81.9%; Quiz 3: 92.3%).

After completion, 92.39% indicated that they would recommend the course to a colleague and 92.20% indicated that they found it useful and informative. Open-ended response options for further thoughts and feedback suggested that the course was generally well received: 'I have worked with female veterans and have found this research to be very relevant to each case. This course has given me more insight into what female veterans face during service and just how much they can be impacted. A very enlightening course'; 'I think everyone who has any contact with women veterans should have access to this very informative course'.

### 3.2. Course 2: Advanced understanding of women veterans – how services can better meet their needs

The 2 h workshop was delivered 14 times, with a total of 363 professionals attending, 238 (65.56%) of whom had previously completed Course 1. The intended target audience of veteran specialist providers and non-specialist healthcare providers was reached, with 82.92% of live session attendees representing these occupational groups. As with Course 1, most attendees were based in the UK (90.35%), with limited numbers from Scotland (2.20%, $n = 8$). One webinar delivered to a Canadian cohort accounted for 8.82% ($n = 32$) attendees from outside the UK.

A small number of attendees (14.2%, $n = 46$) completed the voluntary feedback at the end of the live workshop. When asked whether they found the workshop useful and whether they were likely to apply the knowledge from the workshop in their professional work, the majority agreed (88.10% and 88.09%, respectively). All respondents agreed that the strategies and considerations for engaging and supporting women veterans presented were feasible and practical within their organization or practice, and 90.47% indicated that they would recommend the course to a colleague. Open-ended response options for further thoughts and feedback indicated that the course was generally well received: 'The discussion was interesting especially because there were people from a variety of backgrounds and hearing other points of view and knowledge shared was rewarding'; 'This workshop gave me a better understanding of the needs and experiences of women veterans, which has improved my understanding of what we may need to do to meet the needs of our female veterans'.

Course 2 materials were made available as a self-guided course in response to requests from attendees of the live sessions for continued access to the resources (Table 1).

### 3.3. Course 3: Advanced understanding of women veterans plus cognitive processing therapy (CPT) training – quantitative results

Two cohorts of clinicians were trained, resulting in a total of 98 UK clinicians working across 34 organizations providing UK clinical services trained in CPT (Table 1).

A total of 57 clinicians completed the first CPT training. Most were female (87.72%, $n = 50$) and had over 5 years' experience working with PTSD (71.93%, $n = 47$). Half had at least one woman veteran currently on their caseload (50%, $n = 29$) and six clinicians held supervisory or lead roles within their service. Pre-training, 10.53% ($n = 6$) had used CPT over the previous 3 months, although none had used it to treat PTSD related to MST in women veterans.

Cohort 1 clinicians were followed up 3 months after training; 68.4% ($n = 39$) responded to the survey, but reasons were not specified for those who did not complete the measures. Seventeen clinicians reported using CPT after training, five of whom indicated using CPT to treat PTSD related to MST in women veterans. Of the clinicians who had not used CPT ($n = 22$), most described not having the opportunity to implement CPT with their current caseload as the reason for this ($n = 14$) (see Table 2 for further details).

A Wilcoxon signed-rank test indicated that clinician self-reported confidence was significantly higher 3 months post-training ($Md = 3.00$, $n = 39$) compared to pre-training ($Md = 2.00$, $n = 39$, $z = -4.756$, $p < .001$), with a large effect size ($r = 0.915$). The mean increase in confidence was 1.00 ($SE = 0.12$). None reported a decrease, 12 reported no change, and 27 reported increased confidence.

Owing to the high demand for CPT places, a second cohort of clinicians was trained to further increase capacity. Cohort 2 included 41 clinicians; most were female (73.17%, $n = 30$) and had over 5 years' experience working with PTSD (56.09%, $n = 23$). Over one-third had at least one woman veteran currently on their caseload (39.02%, $n = 16$), and over half of clinicians (51.22%, $n = 21$) held supervisory or lead roles within their service. Pre-training, 19.51% ($n = 8$) had used CPT over the previous 3 months, although only one had used it to treat PTSD related to MST in women veterans. For cohort 2, the eligibility criteria were extended to include clinicians who work with veterans of all genders, (not exclusively women veterans) to further increase UK clinical capacity.

### 3.4. Course 3: Cohort 1 – qualitative results 3 months post-training

Responses from the first cohort of clinicians coalesced around three main themes during analysis of the dataset: (1) Clinician confidence delivering CPT; (2)

**Table 1.** Education programme attendees (individual attendees $n = 684$).

| Total attended | Course 1<br>526 | Course 2 (Live)<br>363 | Course 2 (Self-guided)<br>33 | Course 3<br>98 |
|---|---|---|---|---|
| Location | | | | |
| England | 417 (79.28) | 164 (45.18) | 28 (84.85) | 81 (82.65) |
| Northern Ireland | 15 (2.85) | 9 (2.48) | – | 4 (4.08) |
| Outside the UK | 8 (1.52) | 33 (9.09) | 1 (3.03) | 1[a] (1.02) |
| Scotland | 44 (8.37) | 8 (2.20) | 3 (9.09) | 4 (4.08) |
| UK-wide | – | 101 (27.82) | – | – |
| Wales | 42 (7.98) | 46 (12.67) | 1 (3.03) | 8 (8.17) |
| Unknown | | 2 (0.55) | | |
| Sector | | | | |
| Academia | 16 (3.04%) | 9 (2.48) | 1 (3.03) | – |
| Armed forces charity | 199 (37.83) | 149 (41.05) | 8 (24.24) | – |
| Government | 35 (6.65) | 34 (9.37) | – | – |
| Healthcare | 197 (37.45) | 152 (41.87) | 16 (48.48) | 98 (100) |
| Support/welfare/housing | 55 (10.46) | 12 (3.31) | 6 (18.18) | – |
| Other | 24 (4.57) | 7 (1.93) | 2 (6.06) | – |
| Experience working with PTSD patients (years) | | | | |
| $\leq 5$ | – | – | – | 28 (28.57) |
| 6–10 | – | – | – | 26 (26.53) |
| 11–20 | – | – | – | 26 (26.53) |
| $> 20$ | – | – | – | 13 (13.27) |
| Unknown | – | – | – | 5 (5.10) |
| Age (years) | | | | |
| 18–29 | 52 (9.89) | – | 7 (21.21) | 3 (3.06) |
| 30–39 | 102 (19.39) | – | 6 (18.18) | 30 (30.61) |
| 40–49 | 130 (24.71) | – | 5 (15.15) | 29 (29.59) |
| 50–59 | 171 (32.51) | – | 12 (36.36) | 25 (25.51) |
| $\geq 60$ | 71 (13.50) | – | 3 (9.09) | 10 (10.20) |
| Unknown | – | – | | 1 (1.02) |
| Gender | | | | |
| Woman | | | | 80 (81.63) |
| Man | – | – | | 15 (15.31) |
| Prefer not to answer | – | – | | 2 (2.04) |
| Non-binary | – | – | | 1 (1.02) |

Note: Data are shown as *n* (%).
$n = 158$ Course 2 session attendees bypassed Course 1; $n = 238$ Course 2 session attendees completed both Course 1 and Course 2; all Course 3 attendees completed all courses.
PTSD = post-traumatic stress disorder.
[a]Remote work contract with OpCourage (UK).

Clinicians' perceptions of CPT; and (3) CPT in a service system.

### 3.4.1. Theme 1: Clinician confidence delivering CPT

Confidence using CPT varied among clinicians. Unfamiliarity delivering the therapy and the complexity of managing multiple worksheets meant that some reported a desire for further training before using CPT in practice. Low clinician confidence was implicated as a barrier to implementation, whereas high confidence facilitated implementation.

Factors impacting confidence were reported. Specifically, training and supervision were reported as enhancing clinicians' confidence in delivering the modality. One clinician stated, 'My supervisor is

**Table 2.** Clinicians' experience of using cognitive processing therapy (CPT) 3 months post-training.

| Superordinate theme | Subtheme | Illustrative quote |
|---|---|---|
| Clinician confidence delivering CPT | Clinicians' confidence | 'Chosen to work within a model I feel more confident with' (P1)<br>'I did not feel confident as I don't have a supervisor for CPT' (P20) |
| Clinician perceptions of CPT | Suitable for the patient cohort | 'Veteran engaging well and impact statement has been invaluable' (P16)<br>'Really good approach which I hope to use as soon as an opportunity arises' (P13) |
| | Not suitable for the patient cohort | 'Keeping veterans on track is challenging. The level of written ability in terms of self-expression also varies greatly' (P3)<br>'Better addressed with a more elective modality' (P38) |
| | Wider applicability of CPT to other populations | 'Currently working with 2 male veterans using the CPT protocol' (P10)<br>'1 female veteran sexual violence, 1 firefighter with death-related trauma, 2 × male veterans combat trauma' (P7) |
| CPT in a service system | Lack of resources and opportunities to use CPT | 'We still do not have access to the manuals to aid practice and consolidate learning' (P20)<br>'My current caseload were already on a treatment plan with other treatment methods, e.g. EMDR' (P33) |
| | Embedding CPT into a system | 'Looking to build a pathway and consider group [CPT] intervention as an offer' (P2)<br>'I had disseminated information, brought in clinical development discussions' (P8) |

experienced in using CPT, which helps with my confidence in formulating suitability and eventually implementing the treatment' (P35). Conversely, a lack of supervision was discussed alongside low clinician confidence. Clinicians who had prior training and experience using CPT reported more confidence delivering CPT after completing the 2 day training offered as part of the current project; one clinician had 'used the protocol previously … since the training feel generally more confident and have had good results' (P10).

### 3.4.2. Theme 2: Clinicians' perceptions of CPT

Clinicians' perceptions of CPT varied. These coalesced around three subthemes: (1) clinicians' perception of CPT as suitable for their patients; (2) clinicians' perception of CPT as not suitable for their patients; and (3) the wider applicability of CPT to other patient populations.

*3.4.2.1. Subtheme 1: Clinicians' perception of CPT as suitable for their patients.* Clinicians who had not had an opportunity to implement CPT in the 3 months post-training discussed their intention to use CPT and perceived the modality as suitable for future use. One participant described CPT as a 'Really good approach which I hope to use as soon as an opportunity arises' (P13).

Clinicians who had the opportunity to use CPT in the 3 months post-training discussed the benefits and challenges of the therapy in practice, particularly with veteran groups. Clinicians valued CPT for its effectiveness with shame and guilt: 'When working with guilt and shame-based memories, I really like working with CPT' (P9) and 'Although CPT requires more prep than other trauma-focused protocols, I really enjoy using the model' (P10). Positive patient outcomes were discussed, including improved patient outcome scores and patients getting along well with the therapeutic process.

*3.4.2.2. Subtheme 2: Clinicians' perception of CPT as not suitable for their patients.* Patient-centred challenges to implementing CPT were discussed, including logistical challenges; for example, 'they are not always able to commit to sessions on a regular basis' (P27), and the need for patients to have a certain level of written ability to engage with the worksheets. Other reasons clinicians reported not using CPT included the perception that it was not suited to individuals on their caseload, who were then allocated to alternative treatment modalities. One clinician perceived that their patients' needs were 'Better addressed with a more elective modality' (P38), although the modality in question was not stated.

*3.4.2.3. Subtheme 3: The wider applicability of CPT to other populationsz.* Some clinicians had used CPT to treat women veterans specifically for MST related to PTSD: 'I have been supporting two clients [using CPT] … Both clients are female veterans … experiencing post-traumatic stress (PTSD) … one is dealing with (MST), and the other with domestic violence' (P2). However, CPT training had a wider reach, as clinicians reported using CPT to treat other patient groups including, for example, male veterans and civilian female sexual abuse survivors.

### 3.4.3. Theme 3: CPT in a service system

Clinicians discussed using CPT at the service system level. Two subthemes were identified: (1) lack of resources and opportunities to use CPT; and (2) embedding CPT into a service system.

*3.4.3.1. Subtheme 1: Lack of resources and opportunities to use CPT.* Clinicians identified service system-level barriers as the reason they were unable to implement CPT post-training. Resource constraints hindering CPT delivery included a lack of access to the latest manuals: 'We still do not have access to the manuals to aid practice and consolidate learning' (P20), whereas others reported not having the opportunity to implement CPT with their current caseload. For example, patients who may have benefited from CPT were 'already on a treatment plan with other treatment methods' (P33). Furthermore, some clinicians in lead roles reported having a limited caseload and reduced patient contact, as their responsibilities shifted their focus from direct CPT delivery to clinical supervision and support: '[I have] not worked with a veteran in this time as I have a limited caseload due to leadership responsibilities' (P24).

*3.4.3.2. Subtheme 2: Embedding CPT into a system.* Some clinicians described a ripple effect of the training, leading to wider team development discussions and service system growth in some cases. Knowledge sharing and advocacy for CPT within services was reported; clinicians discussed promoting and disseminating information about CPT during supervision and training, and among professional networks: 'I will shortly be delivering teaching to trainee clinical psychologists about working with military veterans in which I will include information about CPT' (P35). Further organizational investment in training and service developments included colleagues undertaking CPT training to build service pathways; one clinician wrote, 'I have seen an increase in referrals for CPT and the lead psychologist has since completed training herself and is looking at getting funding for the other psychologists in the team to get trained in it' (P32). Clinicians highlighted the value of ongoing training, with both new – 'The training was amazing I would

love to do a refresher at some point too' (P19) – and experienced practitioners noting its importance for maintaining competence and confidence: '[I] found this course extremely valuable … it was good to be able to refresh' (P22).

## 4. Discussion

This paper outlines the three-tiered online education and training programme developed to enhance professional knowledge and improve service support for UK women veterans, specifically those who have experienced MST. Guided by KMb theory, which uses research evidence to inform real-world impact, the programme formed part of a wider KMb initiative and was operationalized in line with the KTA framework (Field et al., 2014; Graham et al., 2006). A mixed-methods evaluation of attendee feedback across all three courses indicated that the programme successfully enhanced professional knowledge. The courses were well received, 98 clinicians were trained in CPT, and the evidence-based service adaptations and considerations proposed in Course 2 were perceived to be practical, relevant, and implementable by those who completed the feedback survey. Positive feedback on Course 2 prompted the research team to convert the live sessions into self-guided materials, which remain available online alongside Course 1, to support sustained knowledge use and growing participation beyond the end of the current project.

### 4.1. KMb frameworks and questions of impact

The KTA framework provided an effective scaffold throughout the project, supporting the interrogation and operationalization of each stage, while maintaining a clear focus on outcomes. The framework linked the early research evidence on UK women veterans through to the education programme. This process involved translating evidence into targeted problems and subsequent solutions while moving from knowledge creation to the action cycle. Stakeholder co-production also ensured that the programme was adapted and implemented in a manner appropriate to the context. However, notable limitations remain.

Phipps et al. (2016) criticise the KTA framework because it does not explicitly define the end goal of the KMb process. They argue that KMb is concerned with sharing knowledge to ultimately support meaningful change and create impact beyond the mere creation of interventions. In their conceptualization, impact can include social, economic, and health benefits, and media and public awareness, as well as addressing specific vulnerabilities. Nonetheless, assessing and measuring what constitutes impact raises several issues. For example, the current study can be viewed as a patient-mediated intervention, as it aims to improve veteran health outcomes indirectly by changing the knowledge and behaviour of health professionals (Stacey & Hill, 2013). However, interventions such as training and clinical capacity building cannot be viewed in isolation outside the context(s) in which they are ultimately delivered and actioned. The professionals who engaged with the education programme worked across different organizations, structures, and contexts; therefore, ongoing monitoring of whether new knowledge was embedded in their daily practice is challenging.

Rutter et al. (2017) argue that rather than concentrating on whether an intervention fixes a problem, the focus should be on whether and how an intervention contributes to reshaping wider systems in a desirable manner. This more holistic approach promotes a move towards continual KMb that 'develops' rather than 'builds' capacity and knowledge by engaging participants at different levels: as individuals, groups, and organizations (Kislov et al., 2014). Viewed through this lens, although participant numbers from Scotland were comparatively low, at less than 10% of attendees across Course 1 ($n = 42$) and Course 2 ($n = 7$), these attendees were predominantly engaged in policy and service design roles at a time when services across the nation were under review and redesign. As such, the impact of the educational programme may be greater than the numbers indicate, and the longer-term impact of the education programme in a Scottish context may yet be realized. The inclusion of international participants may also result in farther-reaching impact, once knowledge has been translated to their specific national contexts. However, assessing this requires in-depth, longitudinal follow-up.

Finally, the KTA framework underplays the role of so-called 'knowledge brokers', whose role is to promote the mobilization of evidence (Davies et al., 2015). In the current study, the research team acted as an initial broker, advocating for and demonstrating evidence-based practice, facilitating the building of professional networks through the live workshops, and recruiting participants. Notably, the decision to continue to make Course 1 and an adapted self-guided Course 2 freely available beyond the stipulated end of the project was made in response to requests from participants who were developing their own KMb projects and wanted to use the materials as signposts, thus assuming the role of brokers.

However, comparatively few participants came from outside veteran-specific support organizations, potentially limiting the project's wider impact. While outreach efforts were made, engagement from veteran employers, for example, remained limited. This highlights a continuing gap in engagement and remains a priority area for future initiatives. Although successful knowledge brokering may have common features irrespective of context (Phipps et al., 2017), moving

beyond one's own disciplinary boundaries into unfamiliar domains such as political or social policy can be challenging (MacKillop et al., 2020). Doing so requires clear guidance in the form of operationalizable KMb frameworks, potentially including more overt benchmarking of project impact.

### 4.2. Implementing and promoting change: CPT clinical capacity building

CPT is a gold-standard treatment, and the structured protocol requires the use of many clinical skills throughout therapy (usually 12 sessions) (Resick et al., 2024). Cohort 1 clinicians who completed CPT training reported significantly higher self-efficacy in delivering CPT 3 months post-training compared to pre-training. Five clinicians had applied CPT to treat PTSD related to MST in women veterans during this period. Although modest, these numbers reflect the early stages of implementation within clinical practice. CPT was also used more broadly across clinicians' varied caseloads, including with male veterans and other blue-light service populations, demonstrating wider clinical impacts.

Notably, the 'events in systems' approach to KMb acknowledges that change requires not just the intervention itself, but also an appreciation that the context in which it is delivered may aid or hinder impact (Skivington et al., 2021). This approach includes wider, interlinked questions around acceptability, implementational considerations, cost-effectiveness, and scalability. Importantly, positive impact is viewed as obtaining information that will be useful to aid decision making and subsequent, future work. In line with this perspective, the current study collected qualitative data exploring clinicians' experiences using CPT, to generate context-sensitive insights into barriers, facilitators, and implementation considerations beyond intervention effectiveness.

Clinicians acknowledged the effectiveness of CPT for addressing feelings of shame and guilt (LoSavio et al., 2023; Resick et al., 2002). Service system-level barriers to implementation included limited opportunities to use CPT and resource limitations. However, those who had not yet had the chance to use CPT expressed their intention to do so in the future, underscoring the importance of longer-term evaluation of evidence-based therapeutic implementation service systems. For individuals who had not encountered service system-level barriers, clinicians' confidence and perception of CPT, factors that are known to shape implementation (Damschroder et al., 2009; LoSavio et al., 2019), impacted its use. Ongoing training and supervision were described as essential. This aligns with work by Pace et al. (2021), who found that post-training consultation and ongoing support improved clinicians' self-efficacy in delivering CPT, which was significantly associated with implementation.

Attendees reported actively disseminating CPT knowledge within their organization, which in some cases led to service system developments and the accommodation of CPT in clinical pathways. Six attendees held clinical lead roles, which limited their opportunities to directly implement CPT owing to smaller caseloads and competing responsibilities, but also enabled them to support other clinicians in delivering CPT. Notably, the second cohort of clinicians trained included a higher number of individuals in supervisory or lead roles. Training individuals in leadership positions has been shown to promote treatment fidelity (the degree of adherence to and competence in delivering a protocol) (Godley et al., 2011). The provision of such training in this study may therefore be considered strategically valuable for increasing service capacity, addressing barriers to implementation (e.g. clinician confidence), and facilitating the organizational adoption of evidence-based interventions.

### 4.3. Implications for future work

Captured within the feedback from attendees on the education programme was a desire for the experiences of men to be included as part of the project. In response, a factsheet on MST was developed and included on the website. It provides an overview of MST in relation to men and women veterans in the UK. Although women veterans may be at higher risk of experiencing MST, owing to the greater number of serving men in the military absolute numbers of those who have experienced such incidents may be comparable (Ministry of Defence, 2025). Underreporting of MST remains an issue irrespective of gender, and future research should look towards strengthening the evidence base on men's experiences, with the aim of improving support for all veterans who have experienced MST.

The authors do not claim that the current project resolves all issues. Rather, they view it as a step in the right direction, whereby the process of evaluating and learning from equivocal findings to stimulate further research can positively generate momentum towards progress (Skivington et al., 2021). The education programme represents a meaningful step towards improving service provision for UK women veterans affected by MST. However, ensuring improved access to effective, evidence-based psychological care for women veterans who have experienced MST undoubtedly extends beyond the scope of a single educational and clinical capacity-building intervention. Achieving this goal requires broad, multifaceted adaptive change across interrelated service systems, encompassing policy makers as well as actors

from the public, private, and third sectors (Rutter et al., 2017).

### 4.4. Strengths and limitations

A key strength of this study is that co-production with women veterans was central to the project, to ensure that the research processes and outputs were informed by lived experience, particularly given the limited evidence base outside the US context. The stakeholder group played a crucial role in informing the education programme and ensuring its relevance to the target population. Although evidence and action from international partners can be transferable, care was taken to adapt the programme's content to the UK context. The inclusion of both UK and international stakeholders represents a key strength of the programme; however, future similar initiatives should continue to ensure that appropriate contextual adaptations are made.

The choice to include true-or-false quizzes to consolidate online learning materials in Course 1 represents a limitation, as evidence suggests that alternative methods such as multiple choice may yield better learning outcomes (Hildenbrand & Wiley, 2025). However, true-or-false options were selected to support user engagement and minimize the time required to complete the online course materials. Future similar programmes may wish to consider alternative consolidation approaches that more effectively balance efficiency with retention.

The iterative nature of this work and the responsiveness to participant feedback have strengthened the relevance and utility of the education programme. Consistent with the KTA framework, this adaptive approach supported the development of a self-guided version of Course 2, and the expansion of available places on CPT training courses for UK clinicians. Eligibility criteria for the second cohort of CPT trainees was broadened to include clinicians working with veterans of any gender, further increasing UK clinical capacity in CPT. The inclusion of the factsheet, alongside the extension of training to clinicians working with veterans of all genders, reflects a recognition that MST can be experienced by any member of the armed forces, irrespective of age, rank, or gender, thus broadening the scope of the programme to enhance awareness for all individuals who may have experienced MST.

Evidence suggests that implementing new training into existing services takes time, and embedding new knowledge or treatment modalities such as CPT within a service system, can take 6–12 months or longer, suggesting that the 3 month follow-up period may be insufficient to assess the long-term impacts on clinical practice or to observe broader service-wide or policy-level changes (Cook et al., 2020; Wiltsey Stirman et al., 2017). Future research should explore the impact of such programmes longitudinally to capture sustained impacts.

More widely, the authors are sensitive to the view that, because the majority of participants were from clinical or support settings, the project could be considered translational research manifesting as continual professional development, rather than KMb, which aims to cast a wider net of influence (Graham et al., 2006). Nonetheless, the efforts to include non-veteran-specific audiences, albeit limited in number, as well as different courses targeting specific groups, can be viewed positively.

## 5. Conclusion

This paper describes a co-produced KMb initiative that successfully operationalized the principles of the KTA framework to enhance professional understanding of and increase support for UK women veterans. The online education programme addressed an important knowledge gap. To date, 684 individuals, primarily professionals working with veterans in the UK, have engaged with the programme, and numbers continue to increase across Courses 1 and 2, which remain available online to sustain knowledge use. Course 3 increased clinical capacity for CPT delivery in the UK, and barriers and facilitators to CPT implementation discussed 3 months post-training aligned with existing literature on clinicians' perceptions of the modality and the importance of ongoing supervision and training when embedding evidence-based therapies in service systems. Ultimately, this work represents a model for translating research into practice and highlights the potential for structured KMb approaches to drive systemic and policy-level change in veteran care. The initiative shows that meaningful impact can be achieved in the short term while simultaneously laying the foundations for sustained, multi-sector progress to better support women veterans in the future.

## Author contributions

CRediT: **Danielle Dryden:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Writing – original draft, Writing – review & editing; **Gavin M. Campbell:** Conceptualization, Writing – original draft, Writing – review & editing; **Dominic Murphy:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing.

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## Data availability

Data are not publicly available, but will be made available upon reasonable request to the corresponding author.

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## Appendix A

**Course 1: Attendee feedback survey**

Unless otherwise specified, responses were scored on a Likert scale from 1 to 5: Strongly disagree (1), Disagree (2), Neutral (3), Agree (4), Strongly agree (5), with higher scores indicating greater agreement.

*Question 1. I found this course useful and informative.*

*Question 2. I would recommend this course to a colleague.*

*Question 3. Please let us know any other thoughts or feedback.*

Free text responses were collected for question 3.

**Course 2: Attendee feedback survey**

Unless otherwise specified, responses were scored on a Likert scale from 1 to 5: Strongly disagree (1), Disagree (2), Neutral (3), Agree (4), Strongly agree (5), with higher scores indicating greater agreement.

*Question 1. I found this workshop useful.*

*Question 2. I am likely to apply the knowledge from this workshop in my professional work.*

*Question 3a. Do you feel the strategies and considerations for engaging and supporting women veterans presented are feasible and practical within your organization or practice?*

Response options for question 3a were Yes or No.

*Question 3b. Please provide a brief explanation for your answer.*

Free text response options were optional for question 3b.

*Question 4a. I would recommend this course to my colleagues/others.*

*Question 4b. Please add any additional comments.*

Free text response options were optional for question 4b.

*Question 5. The workshop content was easy to follow.*

*Question 6. What key learning point will you take away from the workshop?*

Free text response options were optional for question 6.

*Question 7a. Were there any topics discussed that you have liked to explore in more detail?*

Response options for question 7a were Yes or No.

*Question 7b. If so, which ones and why*?

Free text response options were optional for question 7b.

*Question 8. What improvements could be made to improve future workshops?*

Free text response options were optional for question 8.

*Question 9. Do you have any additional comments or feedback?*

Free text response options were optional for question 9.

**Course 3: Trainee survey**

Responses were scored on a Likert scale from 1 to 4, Not at all confident (1), A little confident (2), Quite confident (3), Very confident (4), with higher scores indicating greater self-efficacy using CPT.

*Question 1. How confident are you currently delivering Cognitive Processing Therapy (CPT)?*

*Question 2. Have you used Cognitive Processing Therapy (CPT) over the past three months?*

Response options for question 2 were Yes or No.

*Question 3. Have you used Cognitive Processing Therapy (CPT) over the past three months with women veterans, specifically to treat PTSD related to military sexual trauma (MST)?*

Response options for question 3 were Yes or No.

*Question 4. If you have used CPT over the past three months, please detail your experience. For example, number of patients you use CPT with, CPT patient demographics and how you find delivering the therapy. Otherwise please leave this blank and skip to the next question.*

Free text response options were optional for question 4.

*Question 5. If you have not used CPT over the past three months please describe why, otherwise please leave this blank and skip to the next question.*

Free text response options were optional for question 5.

*Question 6. Have you recommended CPT to colleagues or disseminated this training in any other capacity? If so please describe how.*

Free text response options were optional for question 6.

*Question 7. Have you engaged with any further CPT training over the past three months? If so please describe.*

Free text response options were optional for question 7.

*Question 8. Please enter any other comments or feedback you wish to provide, otherwise please leave this blank.*

Free text response options were optional for question 8.